

ENG-654 · LECTURE 05

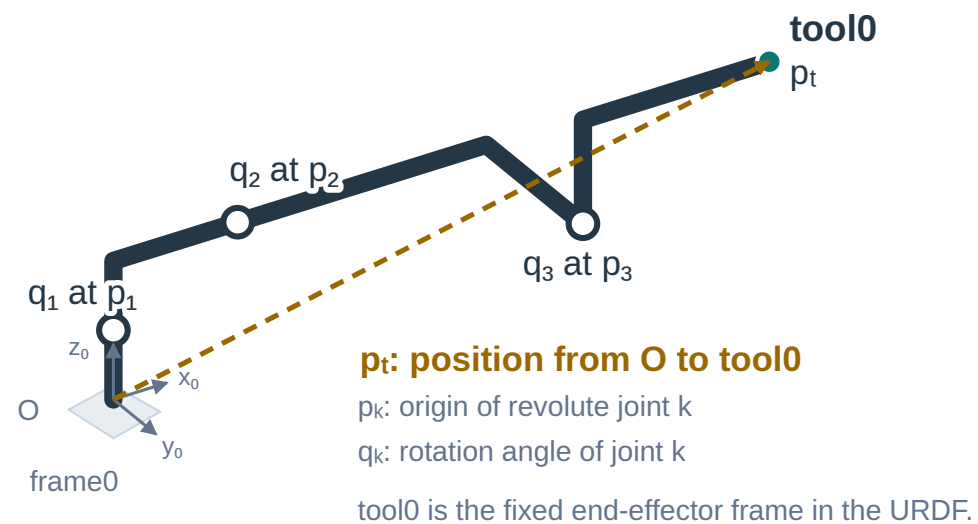
Robot Singularities

From poses to motions: the Jacobian, its determinant, and the paths a robot can follow

Running examples: a spatial 3R robot for end-effector position and the ABB IRB 4600 for full-pose singularities.

Joint angles describe the posture; a point describes the end effector

Robot notation · home posture $\mathbf{q} = (0, 0, 0)$



URDF home posture: $q_1 = q_2 = q_3 = 0$. All vectors use world frame 0.

$\mathbf{q} = [q_1, q_2, q_3]^T$: joint angles [rad]. **Joint velocity** $\dot{\mathbf{q}}$ is their time derivative [rad/s].

$\mathbf{tool0}$ is the **end-effector frame specified in the URDF**, fixed to the final link. Its origin is the point tracked here.

${}^0\mathbf{p}_t$: position of that origin [m]. The subscript t identifies the end-effector point; the left superscript 0 means coordinates in the world frame.

${}^0\mathbf{p}_k$: a point on joint k 's axis. ${}^0\mathbf{z}_k$: the unit direction of that axis, even when the URDF local axis is y .

End-effector linear velocity is ${}^0\dot{\mathbf{p}}_t$ [m/s]. A physical tool need not be attached.

First we study a **position task** (three linear velocity components). Later, a full pose task also includes three angular velocity components.

We know where the robot can be. How can it move?

Lectures 02–04 described poses and solved forward and inverse kinematics. Motion planning connects those poses through time.

So far: a configuration

$$\mathbf{q} \mapsto {}^0\mathbf{p}_t = f(\mathbf{q})$$

Which end-effector position belongs to these joint angles?

Which joint configurations reach a given pose?

Now: a motion

$$\mathbf{q}(t) \mapsto {}^0\mathbf{p}_t(t)$$

Which joint velocities move the end effector along a desired path?

Can the robot maintain that motion at every point?



The missing link is the relation between joint velocity and end-effector linear velocity.

Velocity is the time derivative of position

The position Jacobian ${}^0J_{t,v}(\mathbf{q}) = \partial {}^0\mathbf{p}_t / \partial \mathbf{q}$ maps joint velocity to end-effector linear velocity. Here v identifies the translational rows.

Position

$${}^0\mathbf{p}_t(t) = f(\mathbf{q}(t))$$

Differential

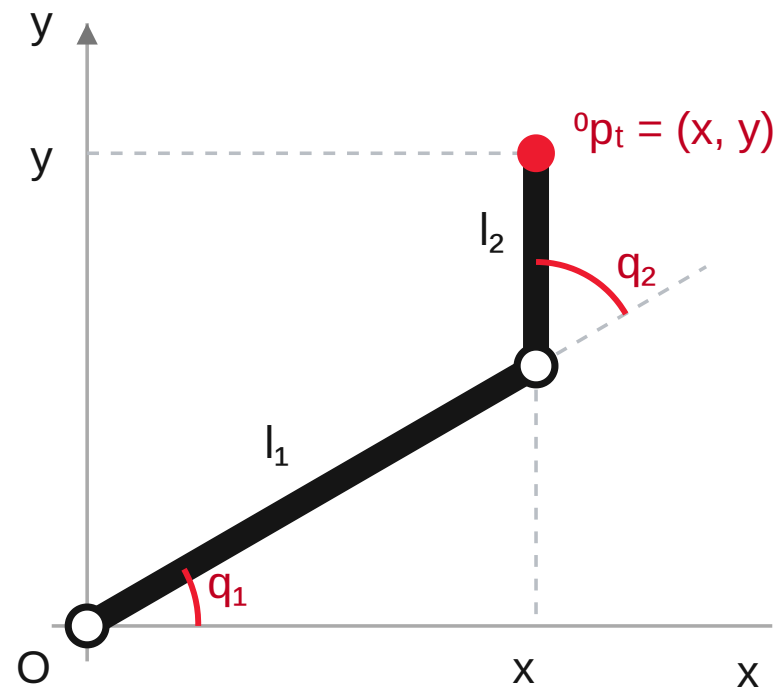
$$d{}^0\mathbf{p}_t = \underbrace{\frac{\partial f}{\partial \mathbf{q}}}_{{}^0J_{t,v}(\mathbf{q})} d\mathbf{q}$$

Chain rule

$${}^0\dot{\mathbf{p}}_t = \frac{d{}^0\mathbf{p}_t}{dt} = {}^0J_{t,v}(\mathbf{q}) \dot{\mathbf{q}}$$

The argument $\mathbf{q}$ is the robot's posture: changing it changes the axes and lever arms. The subscript t identifies the tracked point, not a new argument. Locally, $\Delta {}^0\mathbf{p}_t \approx {}^0J_{t,v}(\mathbf{q}) \Delta \mathbf{q}$ for a small joint displacement. The velocity relation is exact at the current configuration.

Differentiate the two position equations



Angles in radians; link lengths in metres.

Forward kinematics

$$\begin{aligned}x &= l_1 \cos q_1 + l_2 \cos(q_1 + q_2) \\y &= l_1 \sin q_1 + l_2 \sin(q_1 + q_2)\end{aligned}$$

Differentiate in time

$$\begin{aligned}\dot{x} &= -l_1 s_1 \dot{q}_1 - l_2 s_{12}(\dot{q}_1 + \dot{q}_2) \\ \dot{y} &= l_1 c_1 \dot{q}_1 + l_2 c_{12}(\dot{q}_1 + \dot{q}_2)\end{aligned}$$

Collect the joint velocities

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \underbrace{\begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \end{bmatrix}}_{{}^0 J_{t,v}(\mathbf{q})} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}$$

$$s_1 = \sin q_1, \quad c_1 = \cos q_1, \quad s_{12} = \sin(q_1 + q_2), \quad c_{12} = \cos(q_1 + q_2).$$

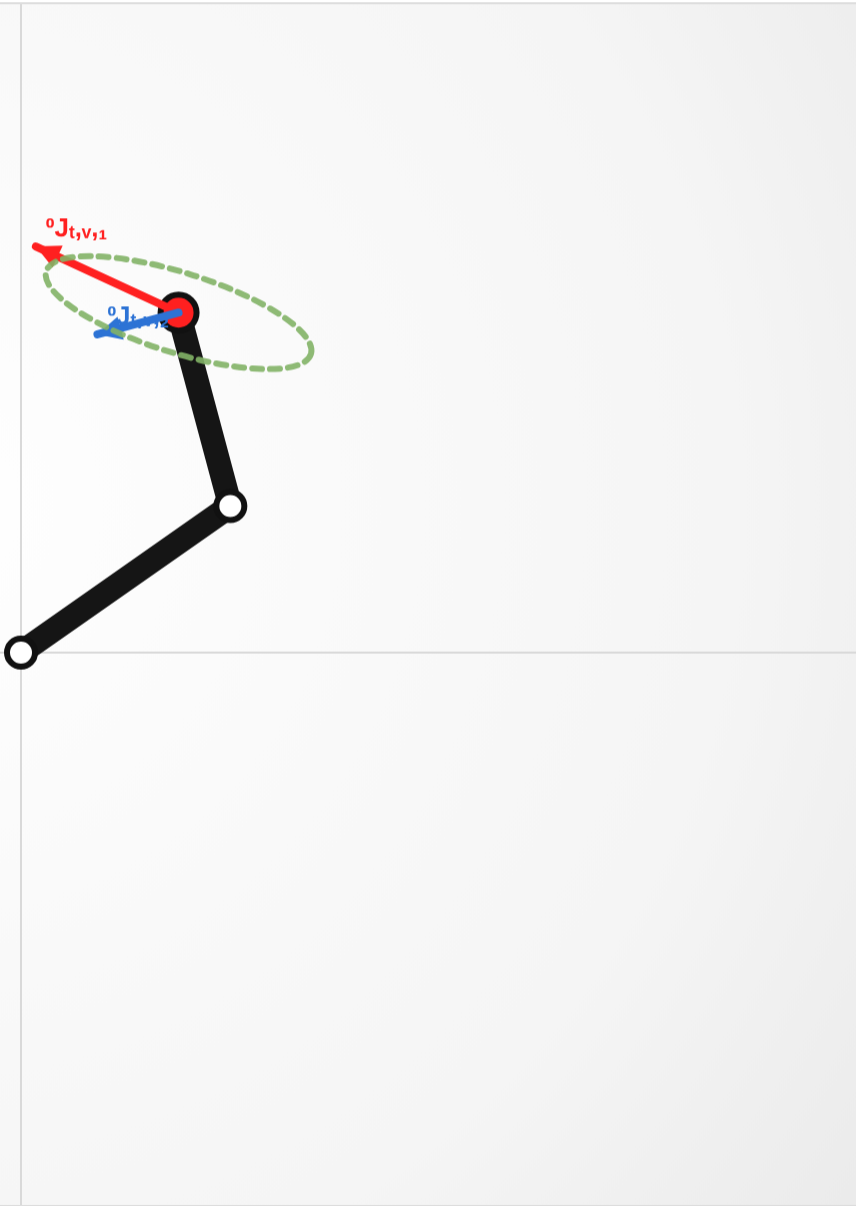
The Jacobian is at the heart of motion planning

Write ${}^0\mathbf{J}_{t,v,k}$ for column k of the position Jacobian. Each column contributes to the end-effector velocity:

$${}^0\dot{\mathbf{p}}_t = {}^0\mathbf{J}_{t,v,1}\dot{q}_1 + {}^0\mathbf{J}_{t,v,2}\dot{q}_2.$$

Try it: change q_2 toward 0° . Watch the two velocity columns become parallel and the velocity ellipse flatten. Can the tool still move in every planar direction?

Drag the red EE · y is up



The diagram shows a 2R planar robot arm in a 2D coordinate system. The base is at the origin. The first joint is at a distance of 1.15 m from the origin, and the second joint is at a distance of 0.90 m from the first joint. The end-effector is marked with a red dot. Two velocity vectors are shown: a red vector labeled ${}^0\mathbf{J}_{t,v,1}$ and a blue vector labeled ${}^0\mathbf{J}_{t,v,2}$. A green dashed ellipse represents the velocity ellipse, centered at the end-effector. The ellipse is elongated along the direction of the red vector and flattened along the direction of the blue vector.

Planar 2R Jacobian

q_1	<input type="range" value="35"/>	35°
q_2	<input type="range" value="70"/>	70°
l_1	<input type="range" value="1.15"/>	1.15 m
l_2	<input type="range" value="0.90"/>	0.90 m

$\det(\mathbf{J})$	0.9726
$ \det(\mathbf{J}) $	0.9726

☒ velocity ellipse

${}^0\mathbf{J}_{t,v} \cdot \text{end-effector linear velocity in world frame } \theta$

[	-1.529	-0.869	]
[	0.709	-0.233	]

Threshold shown: near singular when $|\det \mathbf{J}| < 0.08\text{ m}^2$.

● ${}^0\mathbf{J}_{t,v,1}$ ● ${}^0\mathbf{J}_{t,v,2}$

A Jacobian entry converts one rate into another

$$[J_{ik}] = \frac{[\text{task velocity}_i]}{[\text{joint velocity}_k]}$$

Task component	Revolute velocity [rad/s]	Prismatic velocity [m/s]
Linear velocity [m/s]	m/rad	m/m = 1
Angular velocity [rad/s]	rad/rad = 1	rad/m

Planar 2R position task

Every entry of ${}^0J_{t,v}$ has units m/rad.
 $\det {}^0J_{t,v}$ has units m²/rad².

Full 6R pose task

Linear and angular rows have different units.
With radians treated as dimensionless, $\det J$ has units m³.

Radians are dimensionless in SI; we retain “rad” to show the conversion. Changing metres to millimetres changes determinant magnitudes, but zero remains zero.

At $\det J = 0$, independent motion directions collapse

Expand the 2×2 determinant

$$\begin{aligned}\det {}^0J_{t,v} &= (-l_1 s_1 - l_2 s_{12})l_2 c_{12} + l_2 s_{12}(l_1 c_1 + l_2 c_{12}) \\ &= l_1 l_2 (s_{12} c_1 - s_1 c_{12}) = l_1 l_2 \sin q_2.\end{aligned}$$

For $l_1, l_2 > 0$

$q_2 = 0$ or $\pi \pmod{2\pi}$ makes the links collinear.

The area spanned by the two columns is zero. The velocity ellipse becomes a line.

See the dependence at $q_2 = 0$

$${}^0J_{t,v} = \begin{bmatrix} -\sin q_1 \\ \cos q_1 \end{bmatrix} \begin{bmatrix} l_1 + l_2 & l_2 \end{bmatrix}$$

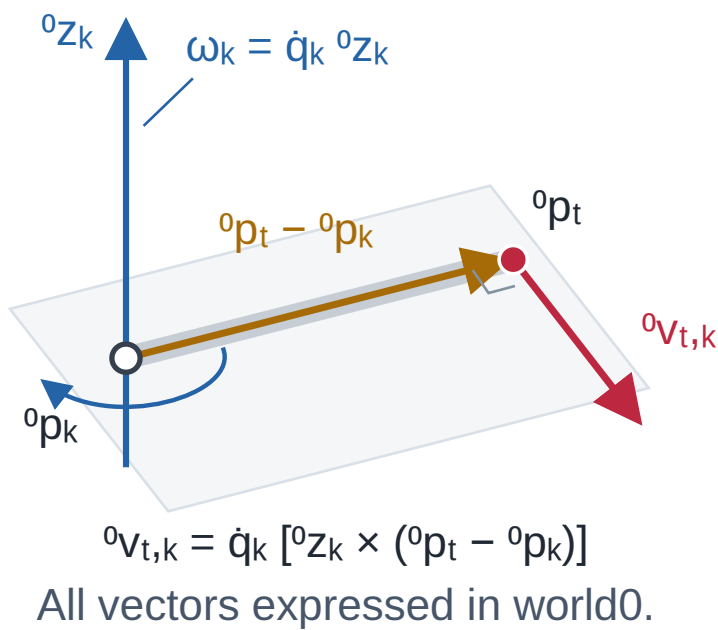
Both columns point along the same tangent: rank 1. Instantaneous radial end-effector motion is unavailable.

For a square task Jacobian, $\det J = 0 \iff \text{rank } J$ is below the task dimension. We now seek this zero set; determinant magnitude alone is not a universal measure of conditioning.

Each column is a familiar screw acting at the tool

World is frame 0. Write ${}^B\mathbf{x}_A$ for entity A expressed in frame B: the frame is a left superscript and the entity is a subscript.

One revolute joint



World-frame screw, angular first

$${}^0\boldsymbol{\xi}_k = \begin{bmatrix} {}^0\mathbf{z}_k \\ {}^0\mathbf{v}_k \end{bmatrix}, \quad {}^0\mathbf{v}_k = -{}^0\mathbf{z}_k \times {}^0\mathbf{p}_k$$

${}^0\mathbf{p}_k$ is a point on joint k's axis. The screw's linear component ${}^0\mathbf{v}_k$ describes its velocity field at the world origin.

Evaluate that motion at the tool

$${}^0\mathbf{J}_{t,v,k} = {}^0\mathbf{v}_k + {}^0\mathbf{z}_k \times {}^0\mathbf{p}_t$$

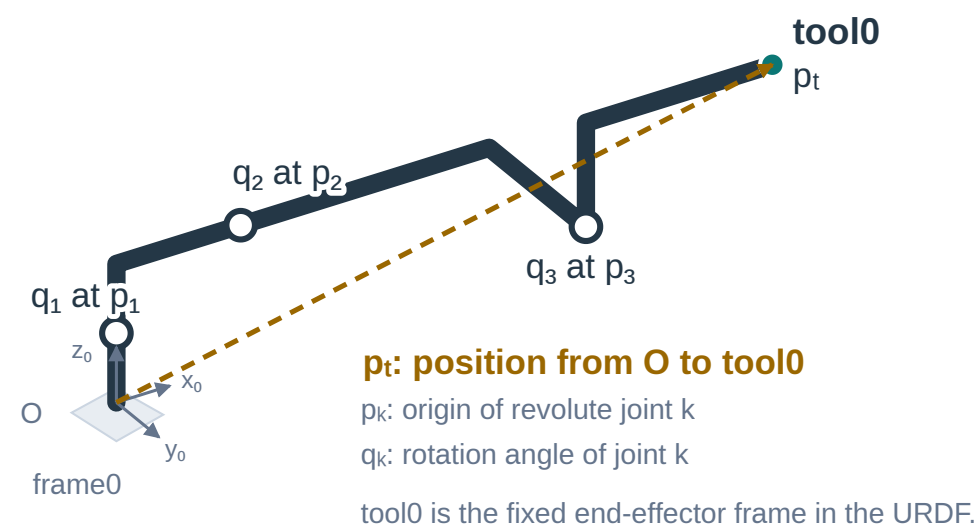
$${}^0\mathbf{J}_{t,v,k} = {}^0\mathbf{z}_k \times ({}^0\mathbf{p}_t - {}^0\mathbf{p}_k)$$

This translational column is the end-effector velocity produced by unit joint velocity.

$$\underbrace{\begin{bmatrix} {}^0\mathbf{z}_k \times ({}^0\mathbf{p}_t - {}^0\mathbf{p}_k) \\ {}^0\mathbf{z}_k \end{bmatrix}}_{\text{revolute, linear first}} \quad \underbrace{\begin{bmatrix} {}^0\mathbf{z}_k \\ \mathbf{0} \end{bmatrix}}_{\text{prismatic, linear first}}$$

Read the joint origins and axes from the URDF

Robot notation · home posture $\mathbf{q} = (0, 0, 0)$



Selected chain

`base_link → link_1 → link_2 → link_3_d3 → tool0`

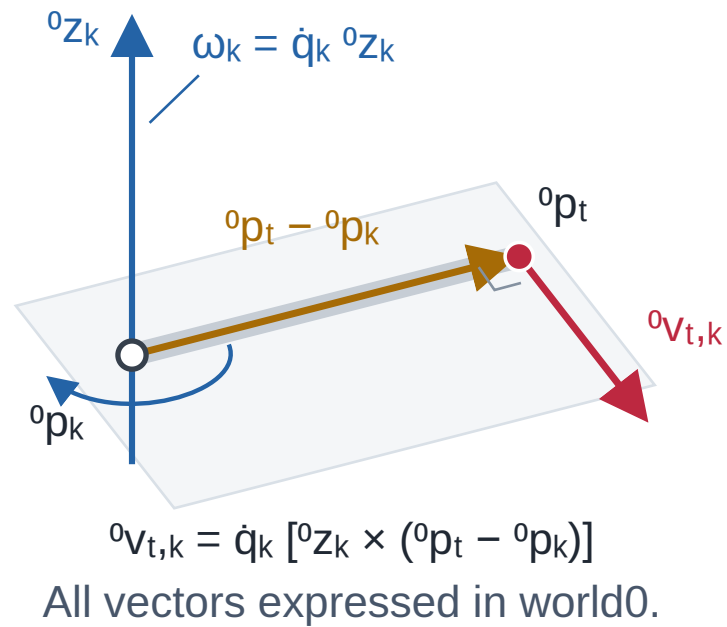
The three joints are revolute. `tool0` is the fixed end-effector frame, not a fourth movable joint.

joint	origin in parent [m]	local axis
1	(0, 0, 0.5)	(0, 0, 1)
2	(1, 0, 0.5)	(0, 1, 0)
3	(2, 1.25, 0)	(0, 0, 1)
tool	(1.5, 0, 0.75)	fixed

Origins are relative to each parent link; all joint-origin rotations are zero. The sketch shows the resulting world positions at $\mathbf{q} = \mathbf{0}$.

One axis and one lever arm give each column

One revolute joint



URDF forward kinematics gives the end-effector position. Let $L = 2 + 1.5c_3$, $B = 1.25 + 1.5s_3$, where $c_i = \cos q_i$, $s_i = \sin q_i$.

$${}^0\mathbf{p}_t = \begin{bmatrix} c_1(1 + Lc_2 + 0.75s_2) - s_1B \\ s_1(1 + Lc_2 + 0.75s_2) + c_1B \\ 1 - Ls_2 + 0.75c_2 \end{bmatrix}$$

$${}^0\mathbf{J}_{t,v,k} = {}^0\mathbf{z}_k \times ({}^0\mathbf{p}_t - {}^0\mathbf{p}_k)$$

$${}^0J_{t,v} = [{}^0\mathbf{J}_{t,v,1} \quad {}^0\mathbf{J}_{t,v,2} \quad {}^0\mathbf{J}_{t,v,3}]$$

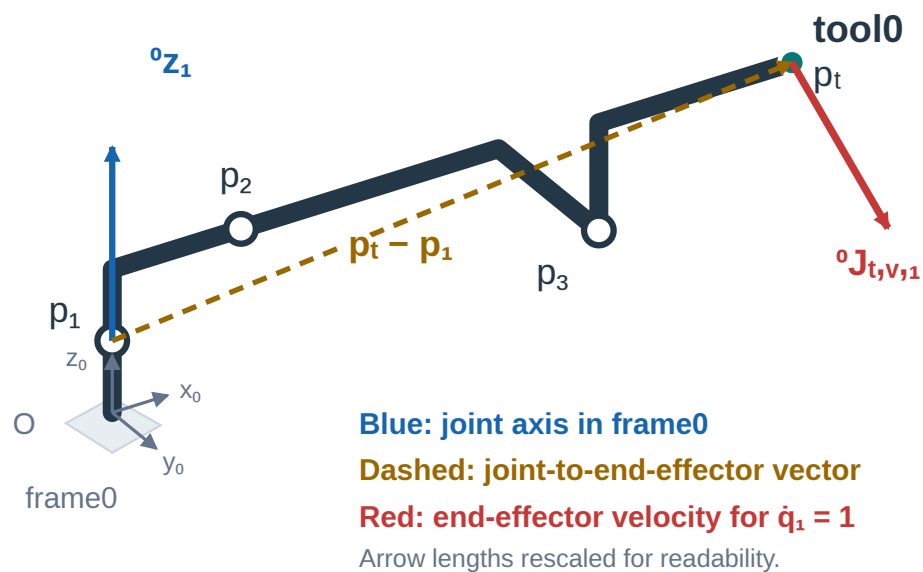
For each joint, locate its axis point ${}^0\mathbf{p}_k$ and direction ${}^0\mathbf{z}_k$, then draw the lever arm to this common end-effector point. All vectors use world coordinates; lengths are metres.

Column k is the end-effector linear velocity for $\dot{q}_k = 1$ rad/s and all other joint velocities zero. The coloured arrows will show this construction one joint at a time.

Joint 1 sweeps the mechanism around its base axis

The picture shows home posture; the equations apply at any $\mathbf{q}$. Use $c_i = \cos q_i$, $s_i = \sin q_i$; lengths are metres.

Joint 1 · same home posture $\mathbf{q} = (0, 0, 0)$



Blue: axis · ochre dashed: lever arm · red: end-effector velocity.

Locate the joint axis in world frame 0

$${}^0\mathbf{p}_1 = \begin{bmatrix} 0 \\ 0 \\ 0.5 \end{bmatrix}, \quad {}^0\mathbf{z}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Take the cross product and expand the column

$${}^0\mathbf{J}_{t,v,1} = {}^0\mathbf{z}_1 \times ({}^0\mathbf{p}_t - {}^0\mathbf{p}_1)$$

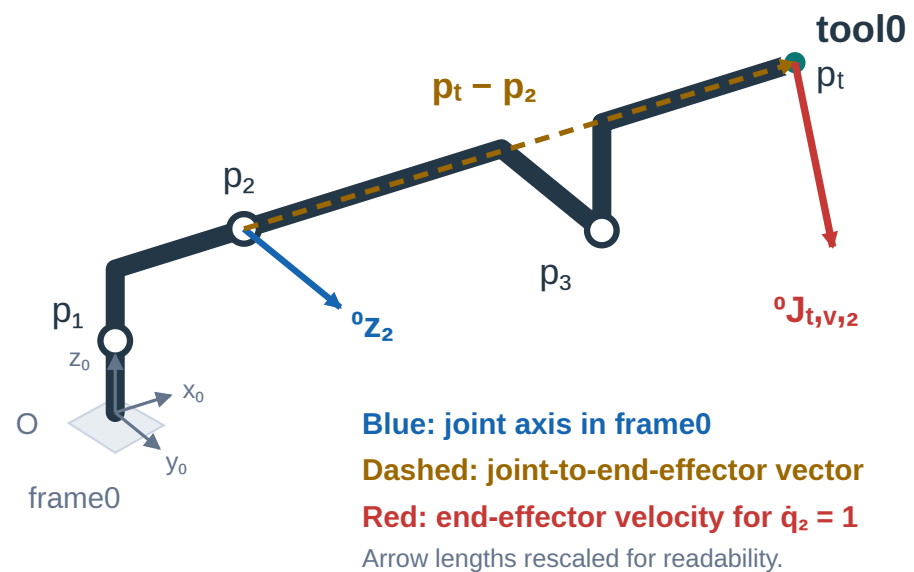
$${}^0\mathbf{J}_{t,v,1} = \begin{bmatrix} -s_1[1 + (2 + 1.5c_3)c_2 + 0.75s_2] - c_1(1.25 + 1.5s_3) \\ c_1[1 + (2 + 1.5c_3)c_2 + 0.75s_2] - s_1(1.25 + 1.5s_3) \\ 0 \end{bmatrix}$$

Check the sketch at $\mathbf{q} = \mathbf{0}$: ${}^0\mathbf{J}_{t,v,1}(\mathbf{0}) = [-1.25, 4.50, 0]^T$ m/rad.

Joint 2 tilts every downstream point about its current axis

The picture shows home posture; the equations apply at any $\mathbf{q}$. Use $c_i = \cos q_i$, $s_i = \sin q_i$; lengths are metres.

Joint 2 · same home posture $\mathbf{q} = (0, 0, 0)$



Blue: axis · ochre dashed: lever arm · red: end-effector velocity.

Locate the joint axis in world frame 0

$${}^0\mathbf{p}_2 = \begin{bmatrix} c_1 \\ s_1 \\ 1 \end{bmatrix}, \quad {}^0\mathbf{z}_2 = \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix}$$

Take the cross product and expand the column

$${}^0\mathbf{J}_{t,v,2} = {}^0\mathbf{z}_2 \times ({}^0\mathbf{p}_t - {}^0\mathbf{p}_2)$$

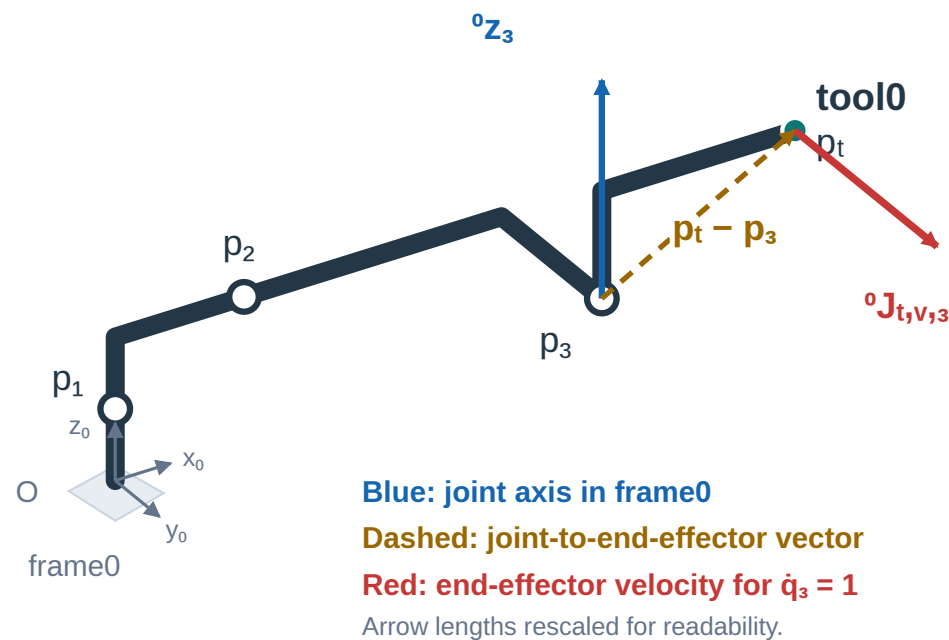
$${}^0\mathbf{J}_{t,v,2} = \begin{bmatrix} c_1[-(2 + 1.5c_3)s_2 + 0.75c_2] \\ s_1[-(2 + 1.5c_3)s_2 + 0.75c_2] \\ -(2 + 1.5c_3)c_2 - 0.75s_2 \end{bmatrix}$$

Check the sketch at $\mathbf{q} = \mathbf{0}$: ${}^0\mathbf{J}_{t,v,2}(\mathbf{0}) = [0.75, 0, -3.50]^T$ m/rad.

Joint 3 moves the end effector through its final lever arm

The picture shows home posture; the equations apply at any $\mathbf{q}$. Use $c_i = \cos q_i$, $s_i = \sin q_i$; lengths are metres.

Joint 3 · same home posture $\mathbf{q} = (0, 0, 0)$



Blue: axis · ochre dashed: lever arm · red: end-effector velocity.

Locate the joint axis in world frame 0

$${}^0\mathbf{p}_3 = \begin{bmatrix} c_1(1 + 2c_2) - 1.25s_1 \\ s_1(1 + 2c_2) + 1.25c_1 \\ 1 - 2s_2 \end{bmatrix}, \quad {}^0\mathbf{z}_3 = \begin{bmatrix} c_1s_2 \\ s_1s_2 \\ c_2 \end{bmatrix}$$

Take the cross product and expand the column

$${}^0\mathbf{J}_{t,v,3} = {}^0\mathbf{z}_3 \times ({}^0\mathbf{p}_t - {}^0\mathbf{p}_3)$$

$${}^0\mathbf{J}_{t,v,3} = \begin{bmatrix} -1.5(c_1c_2s_3 + s_1c_3) \\ 1.5(c_1c_3 - s_1c_2s_3) \\ 1.5s_2s_3 \end{bmatrix}$$

Check the sketch at $\mathbf{q} = \mathbf{0}$: ${}^0\mathbf{J}_{t,v,3}(\mathbf{0}) = [0, 1.50, 0]^T$ m/rad.

Read each axis, lever arm, and velocity directly on the robot

The course STL links follow custom_3R_new.urdf. Every ${}^0\mathbf{z}_k$, ${}^0\mathbf{r}_{t,k} = {}^0\mathbf{p}_t - {}^0\mathbf{p}_k$, and ${}^0\mathbf{J}_{t,v,k} = {}^0\mathbf{z}_k \times {}^0\mathbf{r}_{t,k}$ can be controlled independently.

Try it: activate one axis, its position vector, and its Jacobian column. Check the cross-product direction, then repeat for the other joints.

Reading custom_3R_new.urdf · 4 STL links

☒ Labels

STL opacity 100%

${}^0\mathbf{z}_1$

${}^0\mathbf{z}_2$

${}^0\mathbf{z}_3$

${}^0\mathbf{p}_t \cdot \text{end effector (tool0)}$

${}^0\mathbf{J}_{t,v,1} = {}^0\mathbf{z}_1 \times {}^0\mathbf{r}_{t,1}$

${}^0\mathbf{J}_{t,v,2} = {}^0\mathbf{z}_2 \times {}^0\mathbf{r}_{t,2}$

3R end-effector position Jacobian

q1

0°

q2

0°

q3

0°

view preset

all columns

joint axes

☒ ${}^0\mathbf{z}_1$

☒ ${}^0\mathbf{z}_2$

☒ ${}^0\mathbf{z}_3$

lever arms

☐ ${}^0\mathbf{r}_{t,1}$

☐ ${}^0\mathbf{r}_{t,2}$

☐ ${}^0\mathbf{r}_{t,3}$

Jacobian columns

☒ ${}^0\mathbf{J}_{t,v,1}$

☒ ${}^0\mathbf{J}_{t,v,2}$

☒ ${}^0\mathbf{J}_{t,v,3}$

☒ show STL links

det(J)

-6.56250

${}^0\mathbf{J}_{t,v,1}$

${}^0\mathbf{z}_1 \times {}^0\mathbf{r}_{t,1} \cdot$ tangent to the base-axis sweep

[-1.250, 4.500, 0.000]

${}^0\mathbf{J}_{t,v,2}$

${}^0\mathbf{z}_2 \times {}^0\mathbf{r}_{t,2} \cdot$ end-effector velocity from tilting about joint 2

[0.750, 0.000, -3.500]

${}^0\mathbf{J}_{t,v,3}$

${}^0\mathbf{z}_3 \times {}^0\mathbf{r}_{t,3} \cdot$ end-effector velocity from the final joint offset

[0.000, 1.500, 0.000]

${}^0\mathbf{J}_{t,v}$ · expressed in frame 0

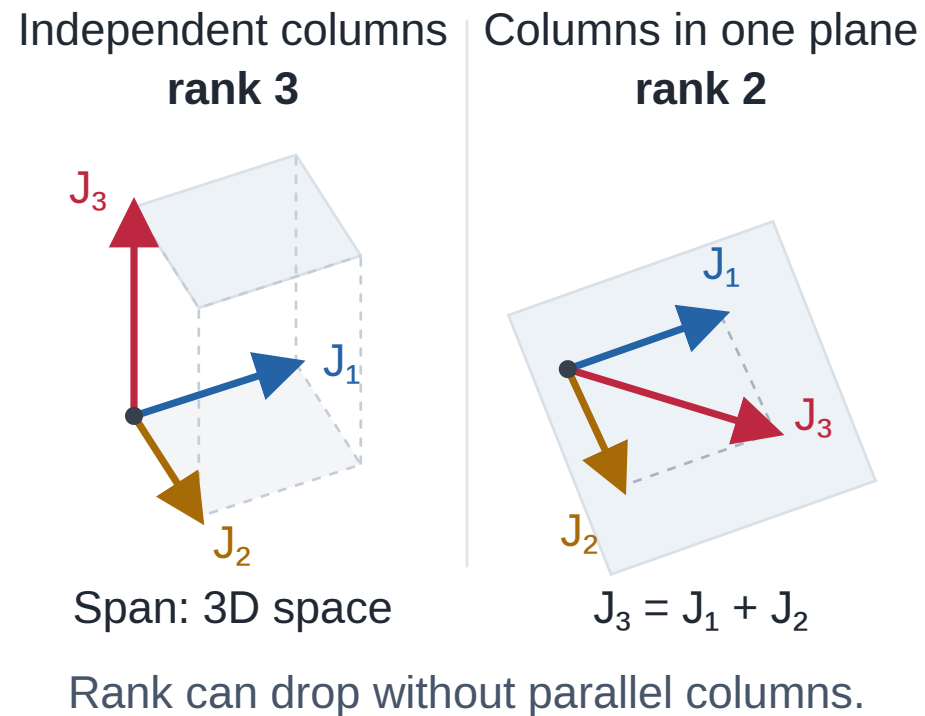
[-1.250 0.750 0.000]

[4.500 0.000 1.500]

custom_3r · rank 3 · toggle ${}^0\mathbf{z}_k$, ${}^0\mathbf{r}_{t,k}$, and ${}^0\mathbf{J}_{t,v,k}$ independently to reconstruct each cross product.

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At a singularity, the three columns are no longer all independent



$$\text{rank } {}^0J_{t,v} < 3 \iff \det {}^0J_{t,v} = 0$$

The rank drops from 3 to 2 or lower. At rank 2, the columns span a plane; they need not all be parallel.

A nonzero joint velocity can give zero end-effector velocity

$${}^0J_{t,v}\dot{\mathbf{q}}_{\text{null}} = \mathbf{0}, \quad \dot{\mathbf{q}}_{\text{null}} \neq \mathbf{0}$$

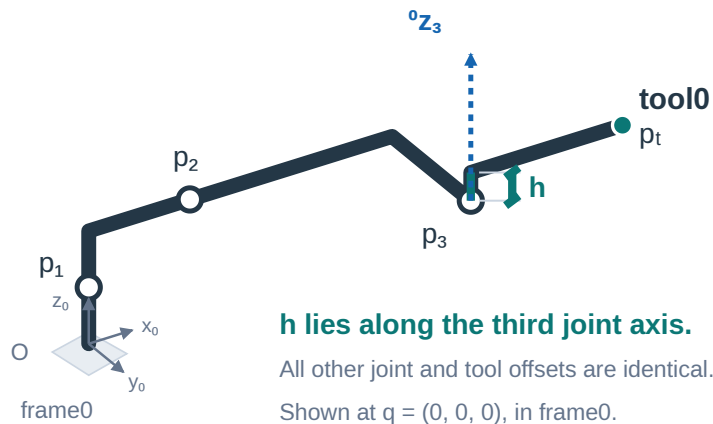
This is an instantaneous joint velocity in the Jacobian's null space. If the rank is r , that null space has dimension $3 - r$.

At least one Cartesian velocity direction is unavailable. Near a singularity, some desired end-effector velocities require very large joint velocities.

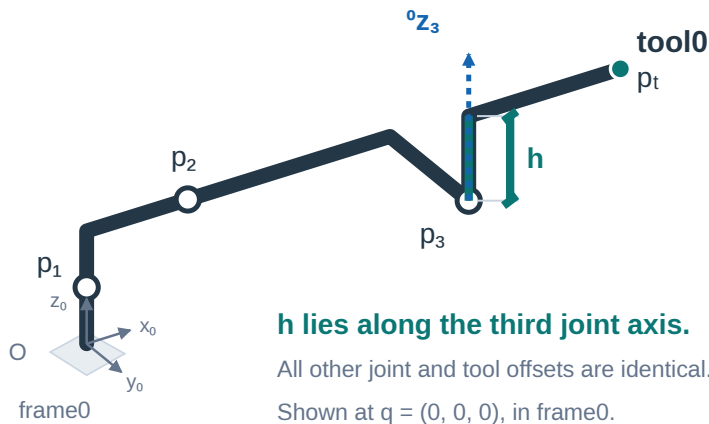
An added end offset changes the lever arms

Same three joint origins and axes, same home posture. Only the fixed displacement to the end-effector point changes.

End offset $h = 0.25$ m



End offset $h = 0.75$ m



No added end offset

The baseline end frame has a fixed axial displacement $h = 0.25$ m from joint 3.

Added end offset: 0.50 m

The end frame now has $h = 0.75$ m. This is the robot used in the preceding column calculations.

Both robots retain the same 1 m offset between the first two axes. Here h is the **end offset along joint 3's axis**; increasing it changes columns 1 and 2, while column 3 stays the same.

The end offset appears in columns 1 and 2

Define $L = 2 + 1.5c_3$, $B = 1.25 + 1.5s_3$, with $c_i = \cos q_i$, $s_i = \sin q_i$. Coloured terms contain h; all lengths are metres.

Baseline end offset: $h = 0.25$ m

$${}^0J_{t,v}|_{h=0.25} = \begin{bmatrix} -s_1(1 + Lc_2 + \mathbf{0.25s_2}) - c_1B & c_1(-Ls_2 + \mathbf{0.25c_2}) & -1.5(c_1c_2s_3 + s_1c_3) \\ c_1(1 + Lc_2 + \mathbf{0.25s_2}) - s_1B & s_1(-Ls_2 + \mathbf{0.25c_2}) & 1.5(c_1c_3 - s_1c_2s_3) \\ 0 & -Lc_2 - \mathbf{0.25s_2} & 1.5s_2s_3 \end{bmatrix}$$

$$\det({}^0J_{t,v}|_{h=0.25}) = 1.5 [Ls_3 + (Lc_2 + \mathbf{0.25s_2})(2s_3 - 1.25c_3)]$$

Added end offset: $h = 0.75$ m

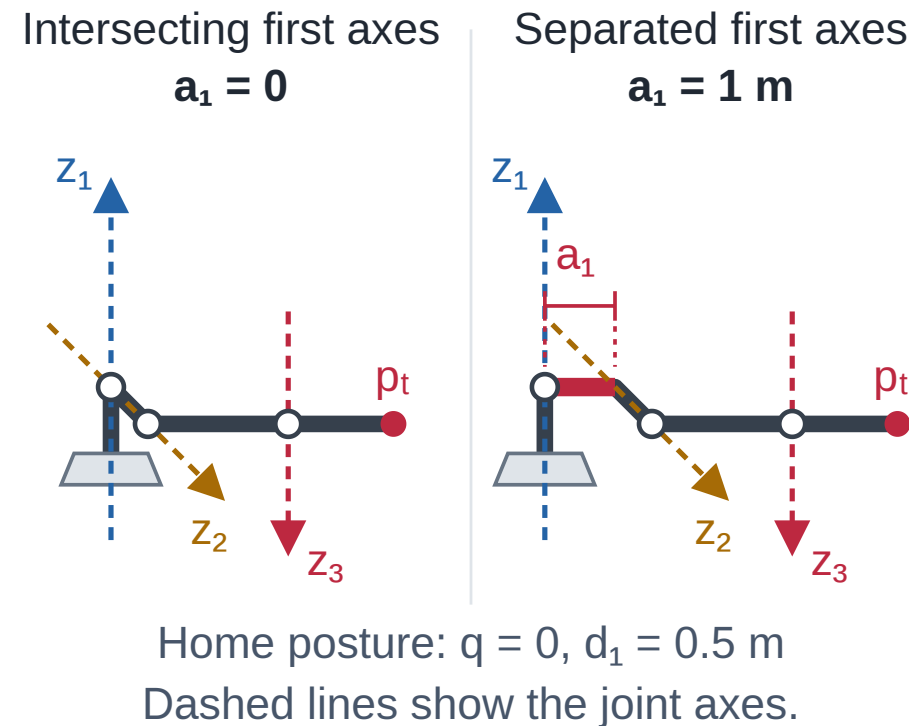
$${}^0J_{t,v}|_{h=0.75} = \begin{bmatrix} -s_1(1 + Lc_2 + \mathbf{0.75s_2}) - c_1B & c_1(-Ls_2 + \mathbf{0.75c_2}) & -1.5(c_1c_2s_3 + s_1c_3) \\ c_1(1 + Lc_2 + \mathbf{0.75s_2}) - s_1B & s_1(-Ls_2 + \mathbf{0.75c_2}) & 1.5(c_1c_3 - s_1c_2s_3) \\ 0 & -Lc_2 - \mathbf{0.75s_2} & 1.5s_2s_3 \end{bmatrix}$$

$$\det({}^0J_{t,v}|_{h=0.75}) = 1.5 [Ls_3 + (Lc_2 + \mathbf{0.75s_2})(2s_3 - 1.25c_3)]$$

Each full matrix comes from $\partial^0\mathbf{p}_t/\partial\mathbf{q}$, equivalently the three axis-lever-arm cross products. Determinants follow from $\mathbf{J}_1 \cdot (\mathbf{J}_2 \times \mathbf{J}_3)$, where $\mathbf{J}_k$ denotes column k of the matrix shown.

Intersecting or separated axes give different singularity sets

For the next comparison and contour map, use a separate D–H family: $a = (a_1, 2, 1.5)$ m, $d = (d_1, 1, 0)$ m, $\alpha = (\pi/2, \pi/2, 0)$. Its end point is the origin after row 3.



No base offset: $a_1 = 0$

$$\det {}^0J_{t,v} = \frac{3}{4}(3c_3 + 4)c_2(c_3 - 2s_3)$$

Two families: $c_2 = 0$ or $c_3 - 2s_3 = 0$.

Base offset: $a_1 = 1$ m

$$\det {}^0J_{t,v} = \frac{3}{4}(3c_3 + 4)[c_2(c_3 - 2s_3) - s_3]$$

The extra term couples q_2 and q_3 . Setting $c_2 = 0$ alone is no longer enough.

$c_i = \cos q_i$, $s_i = \sin q_i$. The sketches use $q = 0$. The height d_1 translates the whole arm and does not affect the Jacobian. The complete matrices follow.

Only column 1 acquires the base-offset contribution

For this D–H family, $L = 2 + 1.5c_3$, $M = 1 + 1.5s_3$, $c_i = \cos q_i$, $s_i = \sin q_i$. Each coloured term is the added a_1 contribution for $a_1 = 1$ m.

No base offset: $a_1 = 0$

$${}^0J_{t,v}|_{a_1=0} = \begin{bmatrix} -s_1Lc_2 + c_1M & -c_1Ls_2 & 1.5(s_1c_3 - c_1c_2s_3) \\ c_1Lc_2 + s_1M & -s_1Ls_2 & -1.5(c_1c_3 + s_1c_2s_3) \\ 0 & Lc_2 & -1.5s_2s_3 \end{bmatrix}$$

$$\det {}^0J_{t,v}|_{a_1=0} = \frac{3}{4}(3c_3 + 4)c_2(c_3 - 2s_3)$$

Base offset: $a_1 = 1$ m

$${}^0J_{t,v}|_{a_1=1} = \begin{bmatrix} -s_1Lc_2 + c_1M - \textcolor{red}{s}_1 & -c_1Ls_2 & 1.5(s_1c_3 - c_1c_2s_3) \\ c_1Lc_2 + s_1M + \textcolor{red}{c}_1 & -s_1Ls_2 & -1.5(c_1c_3 + s_1c_2s_3) \\ 0 & Lc_2 & -1.5s_2s_3 \end{bmatrix}$$

$$\det {}^0J_{t,v}|_{a_1=1} = \frac{3}{4}(3c_3 + 4)[c_2(c_3 - 2s_3) - \textcolor{red}{s}_3]$$

$$\det J = J_{11}(J_{22}J_{33} - J_{23}J_{32}) - J_{12}(J_{21}J_{33} - J_{23}J_{31}) + J_{13}(J_{21}J_{32} - J_{22}J_{31}).$$

Expand along row 1 and collect terms. Since $3c_3 + 4 \geq 1$, only the remaining factor can vanish.

A spatial robot has a singularity set, not one bad posture

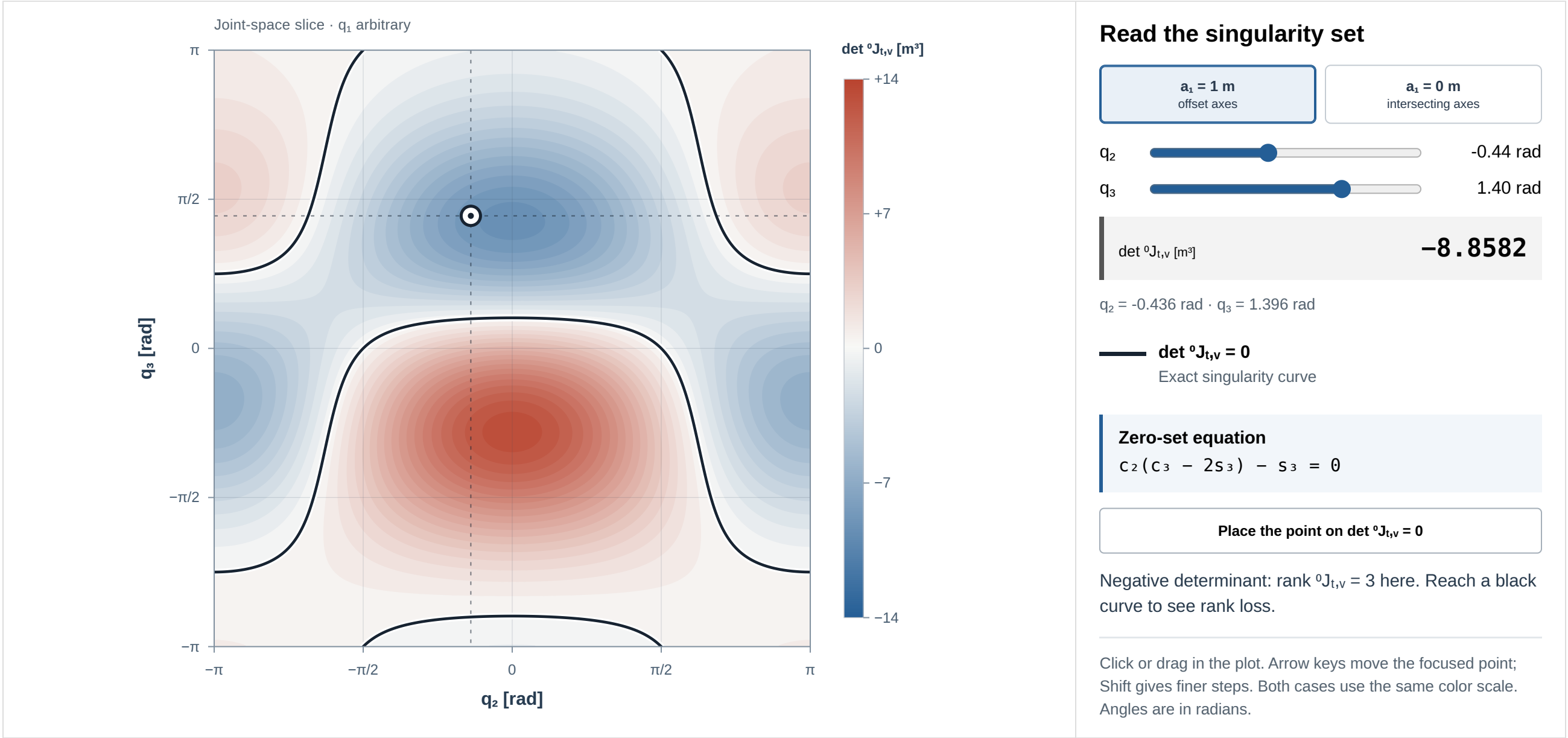
$$\mathcal{S} = \{\mathbf{q} \in \mathcal{C} : D(q_2, q_3) = 0\}, \quad D = \det {}^0J_{t,v}$$

Here $\mathcal{C}$ is the space of joint configurations. With full periodic revolute coordinates, the (q_2, q_3) domain is a torus and the same curves lift across q_1 . Mechanical joint limits clip that torus to a bounded feasible domain.

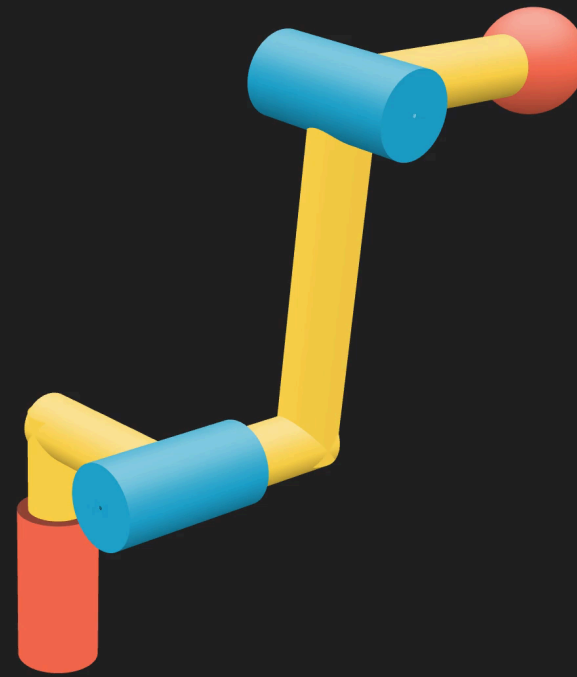
Trace $\det J = 0$ across the joint-angle plane

Color shows the signed determinant. The explicit zero curves mark configurations where the 3R position task loses a motion direction.

Compare: switch a_1 from 1 m to 0. Locate $q_2 = \pm\pi/2$, then inspect the curves and determinant values. Both views use the same color scale.



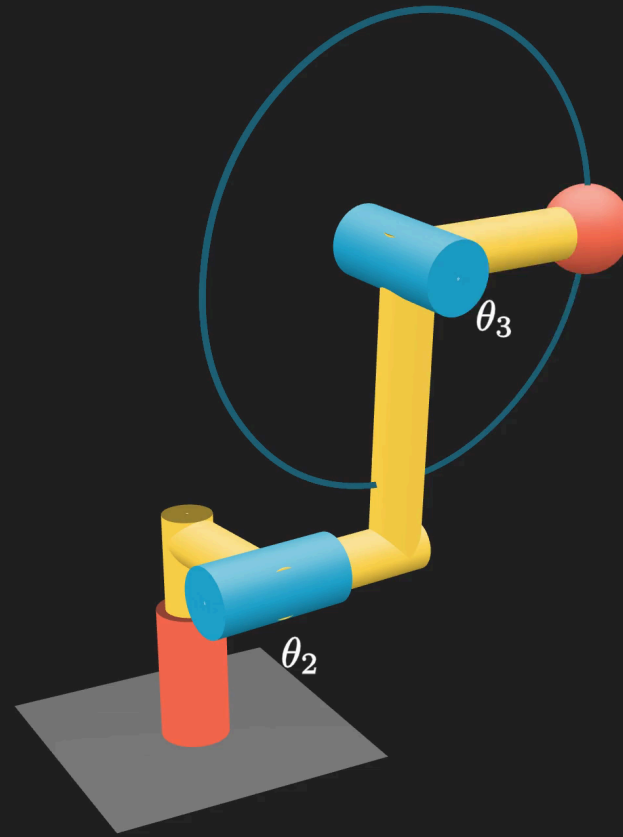
Why the singularity equation does not depend on q_1



An example of a 3R robot

For this 3R family, q_1 rotates the world-expressed velocity columns together, preserving their determinant.

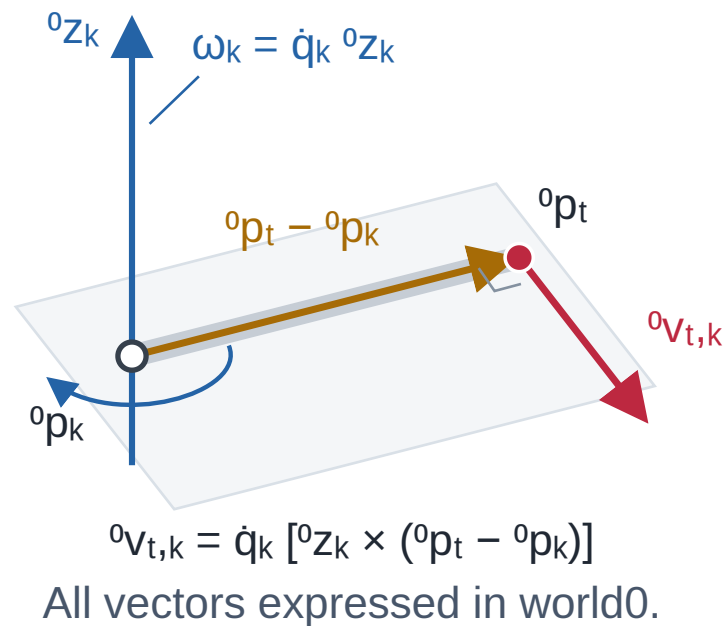
Periodic joint angles form a torus



Identify opposite edges of the periodic (q_2 , q_3) domain. The singularity curves live on this torus; joint limits restrict the feasible domain.

The task selects the velocity components we keep

One revolute joint



Position task · three components

$${}^0\dot{\mathbf{p}}_t = {}^0J_{t,v}\dot{\mathbf{q}}$$

The 3R position Jacobian is 3×3 . It contains only the translational rows.

Full pose task · six components

$$\begin{bmatrix} {}^0\mathbf{v}_t \\ {}^0\boldsymbol{\omega}_t \end{bmatrix} = \underbrace{\begin{bmatrix} {}^0J_{t,v} \\ {}^0J_{t,\omega} \end{bmatrix}}_{{}^0J_t} \dot{\mathbf{q}}$$

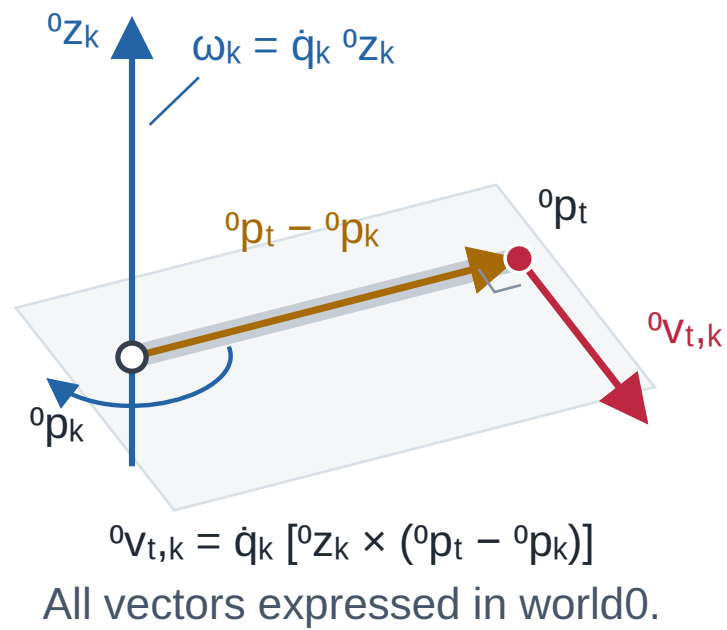
Here ${}^0\mathbf{v}_t = {}^0\dot{\mathbf{p}}_t$. The 6R geometric Jacobian is 6×6 and also includes angular velocity ${}^0\boldsymbol{\omega}_t$.

Both matrices map **joint velocity to the chosen task velocity**. For the ABB example, we will track the wrist center O_5 and write 0J_5 .

An axis and a lever arm determine the point velocity

For joint k , use the world axis ${}^0\mathbf{z}_k$, axis point ${}^0\mathbf{p}_k$, tool point ${}^0\mathbf{p}_t$, and lever arm ${}^0\mathbf{r}_{t,k} = {}^0\mathbf{p}_t - {}^0\mathbf{p}_k$.

One revolute joint



1

$${}^0\boldsymbol{\omega}_{t,k} = \dot{q}_k {}^0\mathbf{z}_k$$

2

$${}^0\mathbf{v}_{t,k} = {}^0\boldsymbol{\omega}_{t,k} \times {}^0\mathbf{r}_{t,k}$$

3

$${}^0\mathbf{v}_{t,k} = \dot{q}_k {}^0\mathbf{z}_k \times ({}^0\mathbf{p}_t - {}^0\mathbf{p}_k)$$

This lecture stacks linear velocity above angular velocity

Change of ordering: in previous lectures, we described screws as angular first, $[\boldsymbol{\omega}; \mathbf{v}]$. We now switch to $[\mathbf{v}; \boldsymbol{\omega}]$. Reordering coordinates preserves the screw's physical motion, axis and pitch, and the rank of a consistently reordered Jacobian.

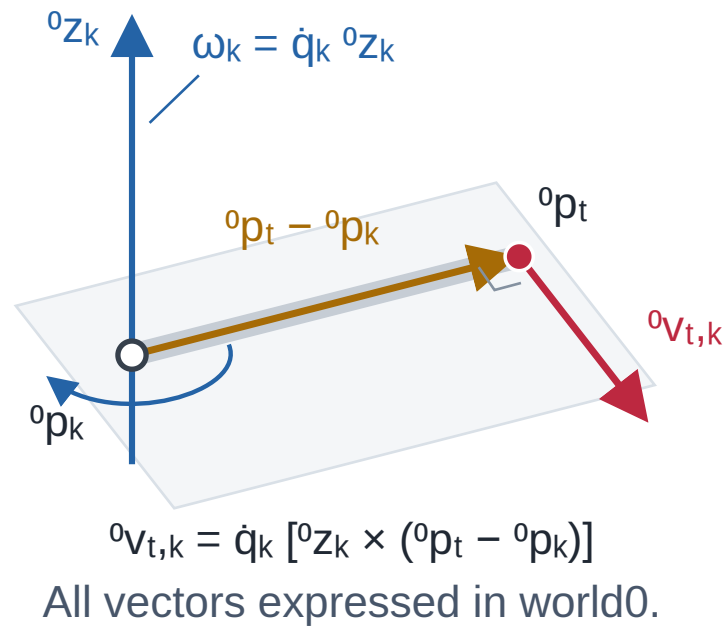
$${}^0\boldsymbol{\xi}_t = \begin{bmatrix} {}^0\mathbf{v}_t \\ {}^0\boldsymbol{\omega}_t \end{bmatrix} = {}^0J_t \dot{\mathbf{q}}$$

$${}^0\mathbf{J}_{t,k} = \begin{bmatrix} {}^0\mathbf{z}_k \times ({}^0\mathbf{p}_t - {}^0\mathbf{p}_k) \\ {}^0\mathbf{z}_k \end{bmatrix}$$

Apply the same permutation to all six-component vectors and transformations. The signed 6×6 determinant changes sign under the row swap; its zero set and rank remain unchanged.

The Jacobian is a catalogue of instantaneous screws

One revolute joint



$${}^0J_t = \begin{bmatrix} {}^0\mathbf{J}_{t,v,1} & {}^0\mathbf{J}_{t,v,2} & \cdots & {}^0\mathbf{J}_{t,v,n} \\ {}^0\mathbf{z}_1 & {}^0\mathbf{z}_2 & \cdots & {}^0\mathbf{z}_n \end{bmatrix}$$

$$\begin{bmatrix} {}^0\mathbf{v}_t \\ {}^0\boldsymbol{\omega}_t \end{bmatrix} = \sum_{k=1}^n {}^0\mathbf{J}_{t,k} \dot{q}_k$$

Shown for revolute joints: each column contains the linear and angular motion per unit joint velocity. Multiplying column k by $\dot{q}_k$ gives that joint's actual contribution. For a prismatic joint, the column is $[{}^0\mathbf{z}_k; \mathbf{0}]$.

The matrix depends on pose coordinates; the physical motion does not

Position-only task

$$\frac{\partial {}^0\mathbf{p}_t}{\partial \mathbf{q}} = {}^0J_{t,v}$$

Differentiating Cartesian position gives exactly the translational geometric Jacobian used for the custom 3R.

Full pose task

$$\mathbf{x} = \begin{bmatrix} {}^0\mathbf{p}_t \\ \phi \end{bmatrix}, \quad \dot{\mathbf{x}} = J_{t,A} \dot{\mathbf{q}}$$

The analytical Jacobian depends on the chosen rotation coordinates: Euler angles, rotation vectors or another representation.

$${}^0\boldsymbol{\omega}_t = E(\phi)\dot{\phi}, \quad {}^0J_{t,G} = \begin{bmatrix} I & 0 \\ 0 & E(\phi) \end{bmatrix} J_{t,A}$$

At a fixed configuration, joint velocities produce the same physical motion. Changing orientation coordinates changes the matrix that reports that motion. We will write the conversion explicitly for roll–pitch–yaw.

One ABB model for the entire 6R derivation

ABB IRB 4600-40/2.55 · [Download the supplied URDF](#). Set world frame 0 = base_link; the URDF's base frame coincides with it.

Read the axis lines at home

Joint	Origin in parent [m]	Local axis
1	(0, 0, 0.495)	z
2	(0.175, 0, 0)	y
3	(0, 0, 1.095)	y
4	(0, 0, 0.175)	x
5	(1.270, 0, 0)	y
6	(0.135, 0, 0)	x

Two useful geometric facts

Axes 2 and 3 are parallel.

Axes 4, 5 and 6 meet at the origin of joint 5: the wrist center O_5 .

All URDF joint-origin rotations are zero. Preserve each supplied joint direction and zero.

The visual DAE files supply the robot's appearance; the URDF supplies its kinematics.

We will derive the D–H table, build the full Jacobian at O_5 in world coordinates, and then simplify its representation.

Keep measured lengths symbolic and joint zeros exact

Use $A_i = R_z(\theta_i)T_z(d_i)T_x(a_i)R_x(\alpha_i)$, with z_{i-1} along joint i. Common normals set a_i ; distances along the joint axes set d_i .

i	θ_i	d_i [m]	a_i [m]	α_i
1	q_1	$d_1 = 0.495$	$a_1 = 0.175$	$-\pi/2$
2	$q_2 - \pi/2$	0	$a_2 = 1.095$	0
3	q_3	0	$a_3 = 0.175$	$-\pi/2$
4	q_4	$d_4 = 1.270$	0	$+\pi/2$
5	q_5	0	0	$-\pi/2$
6	q_6	$d_6 = 0.135$	0	0

Why the joint-2 offset?

At URDF home, the upper arm points along +world z. With $\alpha_1 = -\pi/2$, choosing $\theta_2 = q_2 - \pi/2$ places the positive a_2 direction there.

Six symbolic lengths

Retain a_1 , a_2 , a_3 , d_1 , d_4 and d_6 in the equations. Keep the displayed zeros and fixed right-angle twists to preserve the ABB axis geometry.

D–H frames are constructed frames. Frame 3 means the frame assigned after D–H row 3; it differs from URDF link_3. A fixed ${}^6R_{tool0} = R_z(\pi)$, with zero translation, recovers the supplied tool0 convention. Although a_1 and a_3 both equal 0.175 m here, retain their distinct geometric roles.

Multiply the D–H transforms to obtain ${}^0\mathbf{p}_5$

Let $\psi = q_2 + q_3$, $c_i = \cos q_i$, and $s_i = \sin q_i$. Each A_i is the standard D–H transform from the table.

$${}^0T_5 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 {}^4T_5 = A_1 A_2 A_3 A_4 A_5 = \begin{bmatrix} {}^0R_5 & {}^0\mathbf{p}_5 \\ \mathbf{0}^T & 1 \end{bmatrix}$$

1 · Select the product's last column

$$\begin{bmatrix} {}^0\mathbf{p}_5 \\ 1 \end{bmatrix} = {}^0T_5 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 G \\ s_1 G \\ d_1 + H \\ 1 \end{bmatrix}$$

The origin of frame 5 has coordinates (0, 0, 0) in its own frame. The homogeneous product gives that same point in world frame 0.

2 · Expand the translation entries

$$\begin{aligned} G &= a_1 + a_2 s_2 + a_3 s_\psi + d_4 c_\psi, \\ H &= a_2 c_2 + a_3 c_\psi - d_4 s_\psi. \end{aligned}$$

G collects the signed horizontal offset along the arm's radial direction; $d_1 + H$ is the height. We keep the shoulder factor's name G throughout.

$${}^0\mathbf{e}_r = [c_1, s_1, 0]^T, \quad {}^0\mathbf{e}_t = [-s_1, c_1, 0]^T, \quad {}^0\mathbf{e}_z = [0, 0, 1]^T$$

At home, $\mathbf{O}_5 = (1.445, 0, 1.765)$ m. D–H origins $\mathbf{O}_4$ and $\mathbf{O}_5$ coincide at the spherical wrist center. Wrist rotations leave that point fixed; d_6 places the flange beyond it.

Calculate every world axis by propagating its local direction

Joint k 's axis is ${}^0\mathbf{z}_k$. In the assigned D–H chain it is the z -axis of frame $k - 1$.

Automated D–H extraction

$${}^0T_{k-1} = A_1 \cdots A_{k-1}, \quad {}^0\mathbf{z}_k = {}^0R_{k-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Read the rotation block's third column. Read the translation block for a point on that axis.

Automated URDF extraction

$${}^0\mathbf{z}_k = {}^0R_{J_k} {}^{J_k}\widehat{\mathbf{a}}_k$$

Propagate the URDF origin and preceding joint transforms to joint frame J_k , then rotate its local unit axis. A URDF axis can point along x , y or z .

$${}^0\mathbf{n} = s_\psi {}^0\mathbf{e}_r + c_\psi {}^0\mathbf{e}_z, \quad {}^0\mathbf{k} = c_\psi {}^0\mathbf{e}_r - s_\psi {}^0\mathbf{e}_z$$

$$\begin{aligned} {}^0\mathbf{z}_1 &= {}^0\mathbf{e}_z, & {}^0\mathbf{z}_2 &= {}^0\mathbf{z}_3 = {}^0\mathbf{e}_t, & {}^0\mathbf{z}_4 &= {}^0\mathbf{k}, \\ {}^0\mathbf{z}_5 &= c_4 {}^0\mathbf{e}_t + s_4 {}^0\mathbf{n}, & {}^0\mathbf{z}_6 &= c_5 {}^0\mathbf{k} + s_5 (s_4 {}^0\mathbf{e}_t - c_4 {}^0\mathbf{n}). \end{aligned}$$

Take cross products to calculate the linear Jacobian

Use $U = a_3c_\psi - d_4s_\psi$ and $V = a_3s_\psi + d_4c_\psi$, so $H = a_2c_2 + U$ and $G = a_1 + a_2s_2 + V$.

$${}^0J_{5,v} = \begin{bmatrix} -s_1G & c_1H & c_1U & 0 & 0 & 0 \\ c_1G & s_1H & s_1U & 0 & 0 & 0 \\ 0 & -(a_2s_2 + V) & -V & 0 & 0 & 0 \end{bmatrix}$$

Column 1

Rotates $G {}^0\mathbf{e}_r$ about world z:
 ${}^0\mathbf{J}_{5,v,1} = G {}^0\mathbf{e}_t$.

Columns 2 and 3

Take ${}^0\mathbf{z}_k \times ({}^0\mathbf{p}_5 - {}^0\mathbf{p}_k)$ for $k = 2, 3$. The distinct axis origins give different lever arms.

Columns 4–6

Every wrist axis passes through O₅, so
 ${}^0\mathbf{z}_k \times ({}^0\mathbf{p}_5 - {}^0\mathbf{p}_k) = 0$.

These rows are in m/rad. The matrix has three zero columns because we chose the wrist center as the velocity reference point.

Stack the world angular axes below the linear rows

For readability, define $X = c_5 c_\psi - c_4 s_5 s_\psi$, $Y = s_4 s_5$, and $Z = -c_5 s_\psi - c_4 s_5 c_\psi$.

$${}^0J_5 = \begin{bmatrix} -s_1 G & c_1 H & c_1 U & 0 & 0 & 0 \\ c_1 G & s_1 H & s_1 U & 0 & 0 & 0 \\ 0 & -(a_2 s_2 + V) & -V & 0 & 0 & 0 \\ 0 & -s_1 & -s_1 & c_1 c_\psi & c_1 s_4 s_\psi - s_1 c_4 & c_1 X - s_1 Y \\ 0 & c_1 & c_1 & s_1 c_\psi & s_1 s_4 s_\psi + c_1 c_4 & s_1 X + c_1 Y \\ 1 & 0 & 0 & -s_\psi & s_4 c_\psi & Z \end{bmatrix}$$

$$\begin{bmatrix} {}^0\dot{\mathbf{p}}_5 \\ {}^0\boldsymbol{\omega}_t \end{bmatrix} = {}^0J_5 \dot{\mathbf{q}}, \quad {}^0J_5 \in \mathbb{R}^{6 \times 6}$$

Every entry is now expressed in world coordinates. The upper three rows map joint velocities to wrist-center velocity; the lower three rows are the dimensionless angular axes. Both parts use the same joint order.

The world calculation is correct, but still contains cancellations

The arm translation block is A_0 and the wrist angular block is W_0 . The upper-right zero block gives $\det({}^0J_5) = \det A_0 \det W_0$.

Arm: expand the three linear columns

$$\det A_0 = G [HV - U(a_2 s_2 + V)]$$

Wrist: take the scalar triple product of its axes

$$\det W_0 = {}^0\mathbf{z}_4 \cdot ({}^0\mathbf{z}_5 \times {}^0\mathbf{z}_6) = -\sin q_5$$

$$\det({}^0J_5) = G s_5 [U(a_2 s_2 + V) - HV]$$

H, U and V contain $q_2 + q_3$. Could a better frame expose the remaining factors with less algebra? First, establish why changing this representation preserves the determinant.

At ABB home, the wrist columns already reveal rank loss

$${}^0J_5(\mathbf{0}) = \begin{bmatrix} 0 & 1.270 & 0.175 & 0 & 0 & 0 \\ 1.445 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1.270 & -1.270 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Linear block: rank 3

The first three joints can move O_5 in three independent directions.

Full matrix: rank 5

Columns 4 and 6 coincide: ${}^0J_{5,4} = {}^0J_{5,6}$. Opposite joint velocities cancel.
Consequently, $\det J = 0$.

Lengths are substituted in metres; angles are radians. This is the supplied URDF zero configuration, without changing any joint signs or zeros.

The adjoint's blocks follow the chosen twist ordering

Write $R = {}^0R_i$, so $R^T = {}^iR_0$, and $\mathbf{r} = {}^0\mathbf{r}_{PQ}$. First shift the point, then rotate into frame i.

Previous lectures: angular first

$$\mathcal{A}_{\omega v} = \begin{bmatrix} R^T & 0 \\ -R^T[\mathbf{r}]_{\times} & R^T \end{bmatrix}$$

Acts on $[{}^0\boldsymbol{\omega}_t; {}^0\mathbf{v}_P]$.

This lecture: linear first

$$\mathcal{A}_{v\omega} = \begin{bmatrix} R^T & -R^T[\mathbf{r}]_{\times} \\ 0 & R^T \end{bmatrix}$$

Acts on $[{}^0\mathbf{v}_P; {}^0\boldsymbol{\omega}_t]$.

Reversing the component order swaps the adjoint's block rows and columns. The same rigid-motion transformation is represented in a different ordering.

$${}^iJ_Q = \underbrace{\begin{bmatrix} ({}^0R_i)^T & 0 \\ 0 & ({}^0R_i)^T \end{bmatrix}}_{\det=1} \underbrace{\mathcal{P}({}^0\mathbf{r}_{PQ})}_{\det=1} {}^0J_P$$

For the full square 6×6 geometric Jacobian in a consistent ordering: $\det({}^iJ_Q) = \det({}^0J_P)$ and their ranks agree. Proper rotations and point shifts preserve the determinant.

To form 3J_5 , keep O_5 and rotate into D–H frame 3

The subscript stays 5: the velocity reference remains the wrist center. The superscript changes $0 \rightarrow 3$: both linear and angular components rotate into frame 3.

1 · Calculate 0R_3 from the first three D–H rows

$${}^0R_3 = \begin{bmatrix} c_1 s_\psi & s_1 & c_1 c_\psi \\ s_1 s_\psi & -c_1 & s_1 c_\psi \\ c_\psi & 0 & -s_\psi \end{bmatrix}$$

Its columns are ${}^0\mathbf{n}$, $-{}^0\mathbf{e}_t$, ${}^0\mathbf{k}$. Frame 3 is the D–H frame after the first three joint transforms. These directions follow the arm.

2 · Rotate every column

$${}^3J_5 = \begin{bmatrix} ({}^0R_3)^T & 0 \\ 0 & ({}^0R_3)^T \end{bmatrix} {}^0J_5$$

$P = Q = O_5$, so $r = 0$. This step needs only a rotation of the two component blocks.

Why frame 3? It removes the base rotation q_1 and aligns coordinates with the arm. More entries become zero or simple functions of q_3 , making the determinant easier to factor.

These are absolute velocities expressed in a moving basis. Differentiating the moving-frame position coordinates would require additional transport terms. Frame 3 is the D–H frame derived above.

The same Jacobian becomes sparse in frame 3

$${}^3J_5 = \begin{bmatrix} 0 & a_2s_3 - d_4 & -d_4 & 0 & 0 & 0 \\ -G & 0 & 0 & 0 & 0 & 0 \\ 0 & a_2c_3 + a_3 & a_3 & 0 & 0 & 0 \\ c_\psi & 0 & 0 & 0 & s_4 & -c_4s_5 \\ 0 & -1 & -1 & 0 & -c_4 & -s_4s_5 \\ -s_\psi & 0 & 0 & 1 & 0 & c_5 \end{bmatrix}$$

One column, explicitly rotated

$$({}^0R_3)^T \begin{bmatrix} -s_1G \\ c_1G \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -G \\ 0 \end{bmatrix}$$

Now the determinant is easy to expand

The arm block's second row has one nonzero entry. The wrist block's first column also has one nonzero entry.

$$\det({}^3J_5) = \det({}^0J_5).$$

The wrist-center choice separates arm and wrist factors

$${}^3J_5 = \begin{bmatrix} {}^3J_{5,v}^{\text{arm}} \text{ } \mathbf{A} & \mathbf{0}_{3 \times 3} \\ {}^3J_{5,\omega}^{\text{arm}} \text{ } \mathbf{B} & {}^3J_{5,\omega}^{\text{wrist}} \text{ } \mathbf{W} \end{bmatrix}$$

$$\begin{bmatrix} {}^3\mathbf{v}_5 \\ {}^3\boldsymbol{\omega}_t \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{B} & \mathbf{W} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{q}}_a \\ \dot{\mathbf{q}}_w \end{bmatrix}, \quad \det J = \det A \det W$$

A: arm translation

Joints 1–3 translate O_5 ; B contains their angular contributions.

Zero block

Wrist axes all pass through O_5 , so joints 4–6 cannot translate it.

W: wrist rotation

With O_5 fixed, the last three joints generate orientation changes.

The zero block was created by the choice of point. Frame 3 simplifies the remaining entries. Both choices preserve the full determinant.

A missing wrist translation cannot be repaired by the wrist

$${}^3J_5 = \begin{bmatrix} {}^3J_{5,v}^{\text{arm}} \text{ A} & \mathbf{0}_{3 \times 3} \\ {}^3J_{5,\omega}^{\text{arm}} \text{ B} & {}^3J_{5,\omega}^{\text{wrist}} \text{ W} \end{bmatrix}$$

Shoulder case: $G = 0$

O_5 lies on the base rotation axis. The whole second row of A vanishes:

$$A_{2,:} = [-G, 0, 0] = [0, 0, 0]$$

Translation along the frame 3 y direction is unavailable.

Elbow case: dependent arm columns

When the remaining 2×2 minor is zero, columns 2 and 3 become dependent.

For either case, there is a nonzero $\mathbf{n}$ with $\mathbf{n}^T A = 0$, hence $\mathbf{n}^{T3} \mathbf{v}_5 = 0$.

At $\sin q_5 = 0$, wrist columns lose independence

$${}^3J_5 = \begin{bmatrix} {}^3J_{5,v}^{\text{arm A}} & \mathbf{0}_{3 \times 3} \\ {}^3J_{5,\omega}^{\text{arm B}} & {}^3J_{5,\omega}^{\text{wrist W}} \end{bmatrix}$$

Expand along the first column

$$W = \begin{bmatrix} 0 & s_4 & -c_4 s_5 \\ 0 & -c_4 & -s_4 s_5 \\ 1 & 0 & c_5 \end{bmatrix}$$

$$\det W = -s_4^2 s_5 - c_4^2 s_5 = -s_5$$

Show the full-column dependence

At $q_5 = 0$, ${}^3J_{5,6} = {}^3J_{5,4}$. Opposite rates cancel:

$${}^3J_5 \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & -1 \end{bmatrix}^T = 0$$

At $q_5 = \pi$, the columns are opposite and equal rates cancel. With a regular arm and O_5 held fixed, only two independent angular directions remain.

One 2×2 minor reveals the ABB elbow factor

$$A = \begin{bmatrix} 0 & a_2 s_3 - d_4 & -d_4 \\ -G & 0 & 0 \\ 0 & a_2 c_3 + a_3 & a_3 \end{bmatrix}$$

Expand along row 2, with cofactor sign –

$$\det A = G \begin{vmatrix} a_2 s_3 - d_4 & -d_4 \\ a_2 c_3 + a_3 & a_3 \end{vmatrix}$$

Expand and cancel the opposite $a_3 d_4$ terms

$$\det A = G [a_3(a_2 s_3 - d_4) + d_4(a_2 c_3 + a_3)] = a_2 G(a_3 s_3 + d_4 c_3)$$

The nonzero elbow offset a_3 contributes a sine term. For this ABB, the elbow singularity is $a_3 \sin q_3 + d_4 \cos q_3 = 0$.

The ABB IRB 4600 has three determinant factors

$$\det({}^0J_5) = \det({}^3J_5) = -a_2 \underbrace{(a_3s_3 + d_4c_3)}_F \underbrace{(a_1 + a_2s_2 + a_3s_\psi + d_4c_\psi)}_G \underbrace{s_5}_{\text{wrist}}$$

Elbow: F = 0

$a_3s_3 + d_4c_3 = 0$
Nominal $q_3 \approx -82.154^\circ$ within the supplied joint limits.
Two columns of the arm translation block A become dependent.

Shoulder: G = 0

O_5 reaches the base axis. Rotation of joint 1 then gives no wrist-point translation.

Wrist: $\sin q_5 = 0$

Axes 4 and 6 align. The full Jacobian has dependent wrist columns.

a_2 is a fixed positive length. d_1 and d_6 do not affect this determinant: d_1 shifts the mechanism vertically, and d_6 lies beyond the chosen wrist point. Changing frame simplified J and exposed its factors without changing det J.

The ABB IRB 4600 has three types of singularity

The DAE links follow the supplied URDF. Joint sliders retain its signs and zeros; the three presets isolate the derived elbow, shoulder and wrist factors.

Try it: start with the world-frame matrix at O_5 . Select “Use frame 3 at O_5 ” to reveal the sparse entries. Then change the point and frame independently and compare the determinants.

ABB · world geometry · rank 6/6

☒ Labels

Mesh opacity 100%

O5 · wrist centre

${}^0J_{S1V1,2}$

${}^0J_{S1V1,3}$

${}^0J_{S1V1,2}$

ABB IRB 4600 · three singularity factors

Regular

F = 0 · elbow

G = 0 · shoulder

$\sin q_5 = 0$ · wrist

$F = a_3s_3 + d_4c_3$ [m]
0.93995

$G(q_2,q_3)$ [m]
1.59196

$\sin q_5$ [1]
0.57358

$\det {}^0J_5$ (world) = -0.939814 m³
 $\det {}^0J_5$ (chosen) = -0.939814 m³
 $-a_2 F G \sin q_5 = -0.939814$ m³
rank J: 6/6 · arm: 3/3 · wrist: 3/3

Regular: three arm translations and three wrist rotations give six independent instantaneous motions. Change the frame to expose the zeros in J.

q1 20.00°

q2 15.00°

q3 -35.00°

q4 25.00°

q5 35.00°

q6 -20.00°

World frame at O5

Use frame 3 at O5

0J_5 · at D-H O_5 , expressed in world frame 0

[-0.544 1.557 0.563 0.000 0.000 0.000]

[1.496 0.567 0.205 0.000 0.000 0.000]

[0.000 -1.417 -1.134 0.000 0.000 0.000]

[0.000 -0.342 -0.342 0.883 -0.446 0.807]

[0.000 0.940 0.940 0.321 0.802 0.552]

[1.000 0.000 0.000 0.342 0.397 -0.208]

► Change the point and frame; inspect a column

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Choose the point for zeros and the frame for simple entries

1 · Build 0J_5 geometrically

Read current axes and origins from the ABB chain.
Stack point velocities above angular velocities.

2 · Re-express as 3J_5

Keep O_5 . Rotate both blocks by $({}^0R_3)^T$. The arm matrix becomes sparse.

3 · Factor det J

Two small determinant expansions expose elbow, shoulder and wrist singularities.

$$\det J = -a_2 G(a_3 \sin q_3 + d_4 \cos q_3) \sin q_5$$

The purpose of changing frame is simpler algebra for the same physical motion. It cannot create or remove a singularity.

Write the coordinate-rate conversion explicitly

Choose roll–pitch–yaw $\phi = [\phi, \theta, \psi]^T$ and ${}^0R_t = R_z(\psi)R_y(\theta)R_x(\phi)$, as in the earlier lectures.

Euler rates → world angular velocity

$${}^0\omega_t = E(\phi)\dot{\phi}$$

$$E = \begin{bmatrix} c_\psi c_\theta & -s_\psi & 0 \\ s_\psi c_\theta & c_\psi & 0 \\ -s_\theta & 0 & 1 \end{bmatrix}$$

World angular velocity → Euler rates

$$\dot{\phi} = E^{-1}{}^0\omega_t$$

$$E^{-1} = \begin{bmatrix} c_\psi/c_\theta & s_\psi/c_\theta & 0 \\ -s_\psi & c_\psi & 0 \\ c_\psi \tan \theta & s_\psi \tan \theta & 1 \end{bmatrix}$$

$$T(\phi) = \begin{bmatrix} I & 0 \\ 0 & E^{-1}(\phi) \end{bmatrix}, \quad J_{t,A} = T(\phi) {}^0J_{t,G}$$

$\det E = \cos \theta$. Where $\cos \theta \neq 0$, T is invertible and both Jacobians have the same rank. At $\cos \theta = 0$, the Euler-rate conversion is undefined even if the geometric robot Jacobian is regular.

Here c and s abbreviate cosine and sine of the orientation angles. T is a 6×6 rate-conversion matrix. [Reference: ETH Robot Dynamics, Euler-rate mappings.](#)

Three consistency checks should agree at regular configurations

Finite difference

$$({}^0J_{t,v})_{:,k} \approx [{}^0\mathbf{p}_t(q + \epsilon e_k) - {}^0\mathbf{p}_t(q - \epsilon e_k)] / (2\epsilon)$$

Geometric

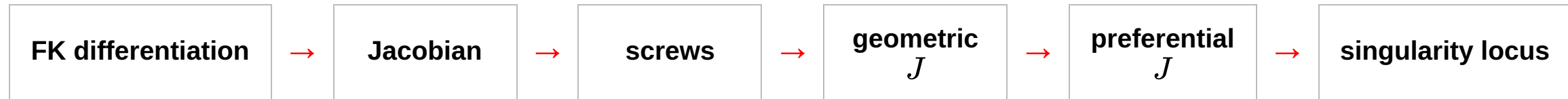
Axis–lever-arm cross products generated from URDF.

Representation

Ordinary and preferential Jacobians preserve rank and determinant zero sets.

If $J \in \mathbb{R}^{6 \times n}$ is non-square, stop saying “determinant”: use rank, $\sigma_{\min}$, or a square task Jacobian.

From a position formula to a singularity equation



1. Singularity is loss of instantaneous task mobility.
2. Each Jacobian column is one joint-generated motion.
3. Changing rigid representation preserves rank.
4. A preferential choice can reveal symbolic structure.
5. A URDF can drive the complete geometric pipeline.

Near-singular paths amplify joint motion before rank is lost

Conditioning matters

As $\sigma_{\min} \rightarrow 0$, some task-velocity directions demand rapidly growing joint velocities.

At $\sigma_{\min} = 0$, the Jacobian loses rank and exact local inversion fails.

$$\sigma_{\min}(J) > 0 : \quad \|J^\dagger\|_2 = \frac{1}{\sigma_{\min}(J)}$$

Path placement

A translated or rotated workspace path can move closer to—or farther from—a singular locus.

Branch continuation

Branches may meet, terminate, or require a coordinated switch at rank loss.

Planning metric

Monitor rank and $\sigma_{\min}$, then enforce joint-velocity, limit, and collision margins.